

慶應義塾大学 SFC 英語 自習プリント

英語が得意な人のための「私大難関本番レベル」— スマートシティの監視ジ
レンマ

2026 年 5 月 26 日

高校 3 年～既卒 (難関私大文系・帰国生・SFC志望)

英語 (慶應 総合政策・環境情報学部・本番水準)

Trillion 塾

第1章 (例題 — 慶應 SFC レベル・5 分)

SFC は 120 分・大問 2 題 (どちらも超長文)。本問は長文 1 題で 50-60 分相当 (SFC の典型ブロック)。

【例題】(慶應義塾大学 SFC 入試 典型形式・改題)

Read the passage and answer the questions that follow.

Beginning in the 2010s, a small number of cities — Singapore, Barcelona, Songdo in South Korea, more recently Toronto and Saudi Arabia's planned megacity NEOM — began to be described, by their planners as much as by their critics, as (A) smart cities. The term referred not to any single technology but to a constellation of interlocking systems: networked sensors that report continuously on traffic and air quality, machine-learning algorithms that anticipate demand for utilities and emergency services, integrated platforms that allow municipal departments to share information that used to live in incompatible silos. The vision was attractive in its broad outlines. A city that could see itself in real time, the argument ran, would be a city that could correct its own malfunctions before they became crises — that could route ambulances around congestion, dispatch buses to where they were actually needed, and detect leaks in water mains long before any human inspector could.

The first decade of serious experimentation has, however, complicated the cheerful initial picture. Some of the complications are technical: sensors fail in unpredicted ways, algorithms turn out to encode the biases of their training data, integrated platforms turn out to be more vulnerable to cyberattack than their designers had hoped. These problems are real, but they are at least the kind of problems engineers know how to talk about. A subtler and harder set of complications has emerged from outside the technical literature, in the work of urban sociologists and political theorists who have begun to ask what kind of city the smart city actually is, and on whose behalf it operates.

The most thoroughly studied case is Toronto's Quayside project, a partnership announced in 2017 between Sidewalk Labs — a subsidiary of Google's parent company — and the local development agency. The Quayside vision was extensive: sensors in every street, lamp post, and rubbish bin; data flowing continuously to a central platform that would optimize

everything from heating to garbage collection; an explicit commitment to make the resulting district a kind of urban research laboratory. Within two years, however, the project had collapsed in the face of sustained public opposition. Residents and civil-society groups pressed a question the original proposal had treated as secondary: who would own the data the city collected, who would set the rules under which it was used, and what recourse would ordinary citizens have if they objected to its uses?

Behind these particular questions lay a more general one. A smart city, by definition, is a city under observation. The same sensors that allow a municipality to detect a water leak also allow it to detect, in principle, where every individual is, with whom they meet, and how long they linger. Whether that capacity is realized depends not on the technology — the technology is the same in either case — but on the legal, political, and corporate constraints within which the technology is deployed. A democracy with a vigorous privacy regime will use the same sensors very differently from an authoritarian state or, for that matter, from a private corporation seeking to monetize the data stream.

The harder question is whether the political constraints that would make smart-city technology safe for democracy can be reliably maintained over the lifetime of the infrastructure. Sensors installed for benign purposes can be repurposed by later administrations under different political pressures; data collected for one use can later be sold or seized for another; promises made by today's planners are not, in general, binding on tomorrow's. A city that has built itself around continuous surveillance has, in effect, given hostages to whatever political configuration comes next.

Some scholars have proposed that smart-city projects be evaluated by a precautionary standard: do not collect data you cannot, in principle, safely permit to be misused. On this standard, many existing proposals would have to be substantially scaled back. Critics of the precautionary view respond that it surrenders too much of the technology's potential benefit, and that the harms it would prevent are speculative while the gains it would forgo are concrete.

The deeper lesson may be that the smart-city question is not, finally, a question about technology. It is a question about the kind of urban life a society wants and the kind of political accountability it is prepared to maintain over the institutions that will increasingly govern that life. The technology gives the question a new urgency, but it cannot, by itself, answer it. The answers will be reached, as they have always been reached in the long

history of cities, by argument among citizens about what their common life ought to look like.

Q1. What does the term "smart cities" refer to, according to the first paragraph?

- ① A single specific information technology used by all cities.
- ② A constellation of interlocking systems — networked sensors, machine-learning algorithms, and integrated municipal platforms — that allow a city to see itself in real time.
- ③ Cities whose residents have above-average intellectual ability.
- ④ Cities that have abolished traditional government.

Q2. According to the second paragraph, what additional complications have emerged beyond the technical problems with smart cities?

- ① Questions raised by urban sociologists and political theorists about what kind of city a smart city is and on whose behalf it operates.
- ② Difficulties with the manufacturing of sensors.
- ③ Increases in the cost of electricity.
- ④ Decline in tourism to smart cities.

Q3. Why did the Toronto Quayside project collapse?

- ① The technology proved physically impossible to install.
- ② Sustained public opposition forced the questions of who would own the collected data, who would set the rules, and what recourse citizens would have, into the foreground.
- ③ A financial crisis bankrupted Sidewalk Labs.
- ④ The Canadian government banned all smart-city projects.

Q4. What general issue lies "behind these particular questions," according to the fourth paragraph?

- ① That the technology itself is too expensive.
- ② That a smart city is, by definition, a city under observation; whether its surveillance capacity is benign depends on the legal, political, and corporate constraints within which the technology is deployed.
- ③ That smart cities consume too much electricity to be sustainable.
- ④ That smart-city projects are illegal under international law.

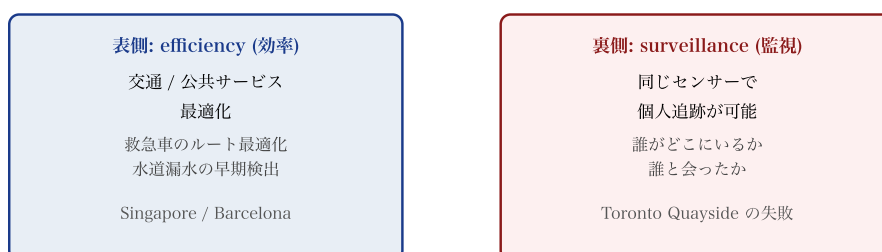
Q5. What is the "harder question" the author identifies in the fifth paragraph?

- ① Whether the political constraints that would make smart-city technology safe for democracy can be reliably maintained over the lifetime of the infrastructure.
- ② Whether sensors can be made waterproof.
- ③ Whether ordinary citizens are intelligent enough to use smart-city services.
- ④ Whether smart cities should pay higher taxes.

Q6. Which of the following best summarizes the author's overall conclusion?

- ① Smart-city technology is a purely technical matter whose problems can be solved by engineers alone.
- ② The smart-city question is, finally, a question about the kind of urban life a society wants and the political accountability it is prepared to maintain; the technology gives the question new urgency but cannot answer it.
- ③ All smart-city projects should be banned without exception.
- ④ Smart-city technology has no real political implications.

スマートシティの双面性 — 効率 ↔ 監視



著者: 「データの所有者」「ルール設定者」を公共の議論で決める必要

図 1-1 スマートシティ — 同じ技術が efficiency と surveillance の両面を持つ

第2章 (解法の全体方針と武器整理・3 分)

SFC エッセイは「技術提示 → 楽観論 → 反証 (技術 + 社会) → 具体例 → 一般化 → 提案 → 政治的含意」の 7 段論法。

【ジャンル分析】

本文は典型的な (A) technology-and-society essay。SFC の入試は(B) 技術と社会の交差点に位置するテーマが頻出 (AI 倫理 / 監視社会 / 気候工学 / ゲノム編集等)。本問は smart city を題材に「技術はそれ単独では答えを出せない」という SFC 的視点を示す。

【本問で必要な 4 つの武器】

- ① (A) 技術的問題 vs 社会的問題の区別 — 第 2 段の「技術的問題は engineer が話せる、社会的問題は別」
- ② (B) 具体例 → 一般化 — Toronto Quayside (第 3 段) → 一般的論点 (第 4 段)
- ③ (C) 技術社会論用語の文脈推測 — "sensor" "algorithm" "surveillance" "precautionary" "infrastructure"
- ④ (D) 著者の立場 = 慎重な技術懐疑論者と読む

【各設問の戦略】

- Q1 (定義) → 第 1 段 "constellation of interlocking systems"
- Q2 (社会的問題) → 第 2 段 "urban sociologists and political theorists"
- Q3 (Quayside 失敗) → 第 3 段 "sustained public opposition ... who would own the data"
- Q4 (一般的問題) → 第 4 段 "a city under observation"
- Q5 (harder question) → 第 5 段 "political constraints ... reliably maintained over the lifetime"
- Q6 (全体結論) → 最終段 "the question is not, finally, a question about technology"

第3章 (Q1 + Q2 — 定義 + 社会的問題・8 分)

「概念定義 + 補足問題の同定」を続けて聞く。

【Q1 設問】 What does the term "smart cities" refer to?

★ 第 1 段の定義

"The term referred not to any single technology but to (A) a constellation of interlocking systems: networked sensors ... machine-learning algorithms ... integrated platforms ..."

"A city that could (B) see itself in real time, the argument ran, would be a city that could correct its own malfunctions before they became crises"

★ Q1 paraphrase

本文 "constellation of interlocking systems: networked sensors, machine-learning algorithms, integrated platforms ... see itself in real time"

選択肢② "a constellation of interlocking systems — networked sensors, machine-learning algorithms, and integrated municipal platforms — that allow a city to see itself in real time"
→ 直接対応。

【Q1 正解】 ② A constellation of interlocking systems — networked sensors, machine-learning algorithms, and integrated municipal platforms — that allow a city to see itself in real time.

【Q2 設問】 What additional complications have emerged beyond technical problems?

★ 第 2 段の論理

"These problems are real, but they are at least the kind of problems engineers know how to talk about. A subtler and harder set of complications has emerged from outside the technical literature, in the work of (A) urban sociologists and political theorists who have begun to ask what kind of city the smart city actually is, and on whose behalf it operates."

★ Q2 paraphrase

本文 "urban sociologists and political theorists ... what kind of city the smart city actually is, and on whose behalf it operates"

選択肢① "Questions raised by urban sociologists and political theorists about what kind of city a smart city is and on whose behalf it operates"

→ 直接対応。

【Q2 正解】 ① Questions raised by urban sociologists and political theorists about what kind of city a smart city is and on whose behalf it operates.

💡 SFC の典型構造: 第 1 段 = 技術的説明、第 2 段 = 「技術的問題」と「社会的問題」の切り分け。 **(A)** 後者がエッセイの本論に入る合図。

第4章 (Q3 + Q4 — Quayside 失敗 + 一般化・10 分)

「具体例 → 一般化」を続けて聞く。SFC の中核設問配置。

【Q3 設問】 Why did the Toronto Quayside project collapse?

★ 第 3 段の論理

"Within two years, however, the project had collapsed in the face of (A) sustained public opposition. Residents and civil-society groups pressed a question the original proposal had treated as secondary: (B) who would own the data the city collected, who would set the rules under which it was used, and what recourse would ordinary citizens have if they objected to its uses?"

★ Q3 paraphrase

本文 "sustained public opposition ... who would own the data ... who would set the rules ... what recourse would ordinary citizens have"

選択肢② "Sustained public opposition forced the questions of who would own the collected data, who would set the rules, and what recourse citizens would have, into the foreground" → 直接対応。

【Q3 正解】 ② Sustained public opposition forced the questions of who would own the collected data, who would set the rules, and what recourse citizens would have, into the foreground.

【Q4 設問】 What general issue lies "behind these particular questions"?

★ 第 4 段の論理

"A smart city, by definition, is (A) a city under observation. The same sensors that allow a municipality to detect a water leak also allow it to detect, in principle, where every individual is, with whom they meet, and how long they linger. Whether that capacity is realized depends not on the technology — the technology is the same in either case — but on (B) the legal, political, and corporate constraints within which the technology is deployed."

★ Q4 paraphrase

本文 "a city under observation ... depends ... on the legal, political, and corporate constraints"

選択肢② "a smart city is, by definition, a city under observation; whether its surveillance capacity is benign depends on the legal, political, and corporate constraints within which the technology is deployed"

→ 直接対応。

【Q4 正解】② That a smart city is, by definition, a city under observation; whether its surveillance capacity is benign depends on the legal, political, and corporate constraints within which the technology is deployed.

★ 重要観察 — Q3 と Q4 の連動

Q3 = 具体例 (Toronto) → Q4 = 一般化 (全 smart city)。 (A) 具体例 → 一般化 は SFC エッセイの基本論法。Toronto の失敗を読んで「一般法則」に登りつめる訓練が SFC の Liberal Arts 教育の核心。

SFC の具体例 → 一般化 → 政治的含意

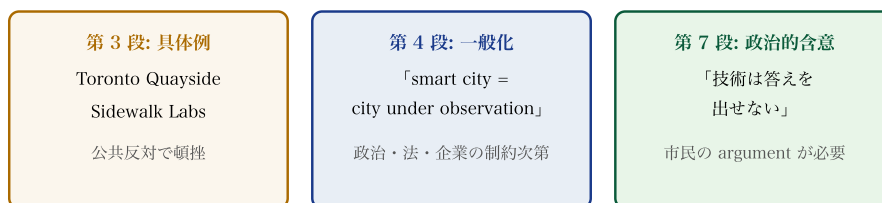


図 5-1 SFC の論理梯子 — 具体例から一般化、そして政治哲学的含意へ

第5章 (Q5 — Harder Question 設問・5 分)

「制約の時間的維持」という第 5 段の核心を読む。

【設問】 What is the "harder question" the author identifies in the fifth paragraph?

★ 第 5 段の論理

"The harder question is whether (A) the political constraints that would make smart-city technology safe for democracy can be reliably maintained over the lifetime of the infrastructure."

具体例:

- "Sensors installed for benign purposes can be repurposed by later administrations"
- "data collected for one use can later be sold or seized for another"
- "promises made by today's planners are not, in general, binding on tomorrow's"

結論: "A city that has built itself around continuous surveillance has, in effect, (B) given hostages to whatever political configuration comes next."

★ paraphrase 対応

本文 "the political constraints ... reliably maintained over the lifetime of the infrastructure"
選択肢① "Whether the political constraints that would make smart-city technology safe for democracy can be reliably maintained over the lifetime of the infrastructure"

→ 直接対応 (ほぼ機械的)。

★ 誤答分析

- ② "sensors can be made waterproof" → (A) 技術的・矮小化。
- ③ "intelligent enough to use" → (B) 無関係。
- ④ "smart cities should pay higher taxes" → (C) 論点ずれ。

【正解】 ① Whether the political constraints that would make smart-city technology safe for democracy can be reliably maintained over the lifetime of the infrastructure.

💡 重要概念: hostages 比喩。スマートシティを作れば、その監視能力は将来のあらゆる政権(民主的とは限らない)に引き継がれる。Snowden 事件後の議論の核心。

第6章 (Q6 — Main Idea 設問 + 関連話題・10 分)

SFC エッセイの結論は「technology cannot answer the question」が王道。

【設問】 Which of the following best summarizes the author's overall conclusion?

★ 最終段の精読

"The deeper lesson may be that (A) the smart-city question is not, finally, a question about technology. It is a question about (B) the kind of urban life a society wants and the kind of political accountability it is prepared to maintain over the institutions that will increasingly govern that life."

"(C) The technology gives the question a new urgency, but it cannot, by itself, answer it.
The answers will be reached, as they have always been reached in the long history of cities,
by (D) argument among citizens about what their common life ought to look like."

→ 主旨 3 要素:

- ①スマートシティ問題は最終的に技術問題ではない
- ②都市生活の理念と政治的説明責任の問題
- ③技術は問いに緊急性を与えるが答えは市民の議論から

★ 選択肢②との対応

"The smart-city question is, finally, a question about the kind of urban life a society wants and the political accountability it is prepared to maintain; the technology gives the question new urgency but cannot answer it."

→ 3 要素すべて含む。

【正解】 ② The smart-city question is, finally, a question about the kind of urban life a society wants and the political accountability it is prepared to maintain; the technology gives the question new urgency but cannot answer it.

★ 関連話題 — SFC 進学後への接続

- ・ (A) Shoshana Zuboff "The Age of Surveillance Capitalism" (2019) — Sidewalk Labs を直接批判
- ・ (B) Bianca Wylie — Quayside 反対運動の中心人物、SFC 系学者がよく引用
- ・ (C) Carlo Ratti (MIT Senseable City Lab) — smart city 楽観派の代表

- ・ (D) Adam Greenfield "Against the Smart City" — 技術懐疑論の代表
- ・ 日本: スマートシティ会津若松 / Toyota Woven City — 本問の論点が日本にも該当

★ 大学進学後への接続

総合政策・環境情報の核心 — 「技術 + 政策 + 社会」の三位一体

1 年次: テクノロジー概論・公共政策論・都市論

2-3 年次: スマートシティ研究会 / プライバシー研究会 / AI 倫理研究会

卒論候補: Toyota Woven City のガバナンス設計 / 大阪万博 2025 のスマートシティ実装 / GDPR と日本の個人情報保護法の比較

第7章 (SFC 英語 攻略のコツ・5 分)

SFC 長文を完答するための 5 つの技術。

【コツ 1】 7 段論法 (技術 → 楽観 → 技術的問題 → 社会的問題 → 具体例 → 一般化 → 政治的含意) を予測

SFC は超長文 (1000-1500 words) なので構造把握が命。「技術提示 → 楽観 → 反証 (2 種) → 具体例 → 一般法則 → 政治的結論」の流れを意識。

【コツ 2】 SFC 頻出の固有名詞・概念

- ・ (A) Sidewalk Labs / Alphabet / Toronto Quayside — 監視資本主義の象徴的事例
- ・ (B) Singapore / Barcelona / Songdo / NEOM — スマートシティの実例
- ・ (C) Shoshana Zuboff / Adam Greenfield / Carlo Ratti — 論者の系譜
- ・ (D) GDPR / Cambridge Analytica / Snowden — プライバシー議論の events

【コツ 3】 SFC 頻出語の文脈推測

sensor / algorithm / surveillance / constellation / municipality / silo / recourse / repurposed / precautionary / accountability

【コツ 4】 誤答パターン 4 種 (SFC 版)

- ・ (A) 技術絶対主義: solved by engineers alone / purely technical — 著者が一貫して批判する立場
- ・ (B) 全面禁止論: should be banned without exception — 著者は中道的
- ・ (C) 政治否定: no real political implications — 本文と正反対
- ・ (D) 具体例の矮小化: waterproof / electricity bills — 一般化失敗

【コツ 5】 時間配分 (SFC 120 分・大問 2 題)

長文 1 題に (A) 50-60 分 を割く。設問数も多い (15-20 問程度) ので時間配分注意。

【最後にひとこと】

SFC の英語は (A) 語学試験 であると同時に (B) 技術社会論リテラシー試験。本問のスマートシティ論は、入学後に学ぶ「技術 + 政策 + 社会」の三位一体的アプローチの予告編。

第8章 (類題チャレンジ・10 分) [10 点]

本講義の手法を応用する 5 問。

(類題 1) 本文中の "silo" の意味として最も近いものを選びなさい。

- ① 部門間で情報が共有されない閉じた管理単位
- ② 農業用の穀物貯蔵塔
- ③ 核ミサイル発射施設
- ④ 高層ビルの建築様式

(類題 2) 本文の "given hostages to whatever political configuration comes next" が示唆するものとして最も適切なものを選びなさい。

- ① 現在作った監視インフラが将来の任意の政権 (民主的とは限らない) に利用可能になる
- ② 外国軍に人質を取られる安全保障上のリスク
- ③ テロリストの脅迫を受ける可能性
- ④ 金融市場の人質経済

(類題 3) 本文中で Sidewalk Labs が引用される目的として最も適切なものを選びなさい。

- ① 公共反対で頓挫した監視主導型スマートシティの代表事例として
- ② 成功した模範事例として
- ③ 本文の著者の所属組織として
- ④ 国際標準化機関として

(類題 4) 本文の "precautionary standard" が示唆する立場として最も適切なものを選びなさい。

- ① 「安全に誤用を許容できないデータは収集しない」という予防原則
- ② 保険会社の標準的リスク評価
- ③ 事前承認制の規制方式
- ④ 輸入品の検疫基準

(類題 5) 本文の論理構成として最も近いものを選びなさい。

- ① 技術提示 → 楽観論 → 技術的反証 → 社会的反証 → 具体例 (Toronto) → 一般化 → harder question (時間的耐久性) → 予防原則の提案 → 政治哲学的結論
- ② 結論先出し → 根拠列举のみ
- ③ 賛成意見の単純な列举
- ④ 歴史的説明のみで結論なし

第9章 (まとめ・宿題・5 分)

今日の核心整理と慶應 SFC 本番に向けた宿題

【本日の核心 5 か条】

- ① SFC エッセイは (A) 超長文 + 7-9 段論法。構造把握が命。
- ② (B) 技術的問題 vs 社会的問題 の切り分けが SFC エッセイの定番技法。
- ③ (C) 具体例 (Toronto/Quayside) → 一般化 (city under observation) → 政治哲学的含意の三段梯子。
- ④ "hostages to ... comes next" のような (D) 政治的時間軸の比喻 を読む力が問われる。
- ⑤ 主旨設問は (A) 「技術は答えを出せない」型結論 を選ぶ。「purely technical / no political implications」は罠。

【宿題】

- ① 本講義の Q1-Q6 を見ないで本文だけ読み直し、自分で 6 問の設問を作成。
- ② 類題 5 問を解き直し、本文 7 段の各段にひと言タイトルを書き出す。
- ③ 過去問: 慶應義塾大学 SFC 英語 (過去 5 年分) から長文 1 題を選び、本日の手法で精読する。

【次回授業】 SFC の英作文 (400-500 words 程度) — "Should Toyota Woven City prioritize efficiency or privacy?" の論理構成

【最後にひとこと】

SFC の英語は (A) 語学試験 であると同時に (B) 技術社会論リテラシー試験。本問のスマートシティ論は、入学後に学ぶ「技術 + 政策 + 社会」の三位一体的アプローチの予告編。

第10章 (発展 — 日本のスマートシティ実装と SFC の役割・5 分)

本問の論点が日本のスマートシティ計画にどう響くかを考える発展編。

【発展題】 SFC で学ぶ「日本のスマートシティ」

★ Toyota Woven City (静岡県裾野市)

- ・ 2021 年着工、富士山麓 70 万平方メートル
- ・ 自動運転車・水素エネルギー・センサーネットワーク
- ・ 「実証実験都市」と位置づけられ、(A) 住民は同意してデータ提供するという建付け
- ・ 本文の論点に照らすと：「同意は撤回可能か」「次世代住民は縛られるか」が SFC の研究対象

★ スマートシティ会津若松

- ・ 2013 年～、Accenture とパートナーシップ
- ・ 市民 ID システムによる住民データ統合
- ・ 成果: 市内サービス利用増、Quayside 型の反対は出ていない
- ・ 本問の論点: なぜ Toronto は失敗し会津は成功したか? = ガバナンスの差

★ スーパーシティ構想 (内閣府)

- ・ 2020 年「国家戦略特区法」改正、データ連携を法的に支援
- ・ 大阪 (2025 万博地区)・つくば・加賀市が認定
- ・ SFC 系研究者が政策設計に関与

★ SFC の研究者の役割

総合政策・環境情報の卒業生は、こうした「技術 + 政策 + 社会」プロジェクトの設計現場に多数。

本問のような英文を読める = 世界の議論を Tokyo の設計に持ち込める = 卒業後の市場価値が高い。

【最終メッセージ】

SFC の英語は単なる読解力ではなく、(A) 「世界の最新議論を日本の現場に翻訳する力」を測る試験。本問の Quayside 失敗を知る学生は、日本の Toyota Woven City をより深く批判的に考察できる。これが SFC で求められる思考の質。